Lab 8: Flip-flops and Counter

Objective: To implement a RS latch and a binary counter with D-type flip flops.

Provided: 74X374 (D flip-flops), 7400 (quad 2-input NAND), 7404 (NOT), 7408 (AND), 7432 (OR), breadboard.

8.1 Experiment: RS Latch

Build an SR latch using 2 NAND gates (Figure 5-4(a), page 250, 6th Edition of Mano and Ciletti book). Demonstrate your circuit to the instructor. Do your observations agree with what you expect (Figure 5-4(b))?
Explain why the input S=0 R=0 should be forbidden.

8.2 Experiment: Binary Counter

Deisgn a binary counter that counts from 0 to 5 then back to 0 and repeat, display the count on LEDs.

1. Examine the pin-out diagram of the 74X374.

```
74374
8-bit 3-state D flip-flop.
    +---+--+---+           +---+---+---*---+
/OE |1   +--+ 20| VCC       |/OE|CLK| D | Q |
 Q1 |2        19| Q8        +===+===+===*===+
 D1 |3        18| D8        | 1 | X | X | Z |
 D2 |4        17| D7        | 0 | / | 0 | 0 |
 Q2 |5   74   16| Q7        | 0 | / | 1 | 1 |
 Q3 |6  374   15| Q6        | 0 |!/ | X | - |
 D3 |7        14| D6        +---+---+---*---+
 D4 |8        13| D5
 Q4 |9        12| Q5
GND |10       11| CLK
    +----------+
```

2. Start by filling the following table:

			D_3	D_2	D_1
$Q_3(t)$	$Q_2(t)$	$Q_1(t)$	$Q_3(t+1)$	$Q_2(t+1)$	$Q_1(t+1)$
0	0	0	0	0	1

3. Next, draw Karnaugh maps to find boolean expressions for each D input in terms of $Q_3(t)$, $Q_2(t)$, and $Q_1(t)$. Mark the inputs from 110 and 111 as don't care conditions in the state table and K-maps.

4. Connect the 4-bit output of the counter to the LEDs.